KIIT DU • SAP Integration Developer Program
Batch: 2027_SAP(OE)

SAP CAPSTONE PROJECT REPORT

SAP Enterprise Blueprint: End-to-End Configuration of a Pharmaceutical Manufacturing Company

Company: ZenithPharma India Pvt. Ltd.

Industry: Pharmaceutical Manufacturing & Distribution

Submitted By

Name: Tushar Raj Padhy
Roll Number: 23052851
Program: SAP Integration Developer
Batch: 2027_SAP(OE)

Submission Date: April 21, 2026 | KIIT DU

1. Project Title & Problem Statement

1.1 Project Title

SAP Enterprise Blueprint — Configuration of ZenithPharma India Pvt. Ltd. across Financial Accounting (FI), Materials Management (MM), and Sales & Distribution (SD) modules of SAP ECC 6.0 / S/4HANA.

1.2 Problem Statement

The Indian pharmaceutical sector is among the most regulated and complex industries globally. A mid-sized pharma manufacturer must simultaneously manage batch-controlled raw material procurement, strict quality-compliance inventory, government-mandated pricing structures (DPCO — Drug Price Control Order), GST-compliant invoicing, and multi-channel distribution across wholesale, retail, and hospital segments. Managing these operations through disconnected legacy systems results in:

- Inability to trace raw material batches to finished drug batches — a critical regulatory requirement under Schedule M of the Drugs and Cosmetics Act.
- Manual reconciliation between procurement records and financial books, leading to audit failures and delayed month-end closing.
- Inaccurate inventory valuation due to multiple costing methods applied inconsistently across warehouses.
- Delayed billing cycles and poor accounts receivable management leading to cash flow problems.
- Lack of real-time visibility into plant-wise production costs and profitability by product line.

This Capstone Project presents a detailed SAP configuration blueprint for ZenithPharma India Pvt. Ltd. — a fictitious pharmaceutical manufacturing and distribution company — addressing all the above challenges through a structured, integrated SAP implementation covering the FI, MM, and SD functional modules.

2. Company Profile — ZenithPharma India Pvt. Ltd.

2.1 About the Company

ZenithPharma India Pvt. Ltd. is a fictitious mid-sized pharmaceutical manufacturing and distribution company incorporated under the Companies Act, 2013. The company is headquartered in Hyderabad, Telangana, and operates two GMP-certified manufacturing plants — one in Hyderabad and one in Ahmedabad. ZenithPharma produces a range of formulations including tablets, capsules, syrups, and injectables, and distributes them through a network of wholesale distributors, hospital chains, and direct retail pharmacies across India and select export markets.

2.2 Company Overview

Field	Details
Company Name	ZenithPharma India Pvt. Ltd.
Industry Sector	Pharmaceutical Manufacturing & Distribution
Headquarters	Hyderabad, Telangana, India
Manufacturing	Hyderabad Plant (ZP01), Ahmedabad Plant (ZP02)
Founded	2004
Annual Turnover	INR 620 Crores (approx.)
Employees	~2,100
Regulatory Status	GMP Certified | WHO-GMP | ISO 9001:2015
Product Lines	Tablets, Capsules, Syrups, Injectables, APIs
Markets Served	Domestic (PAN India) + Exports (South-East Asia, Africa)
Currency	Indian Rupee (INR)
Fiscal Year	April 1 to March 31 (Indian Fiscal Year)
SAP Platform	SAP ECC 6.0 (with planned S/4HANA migration)

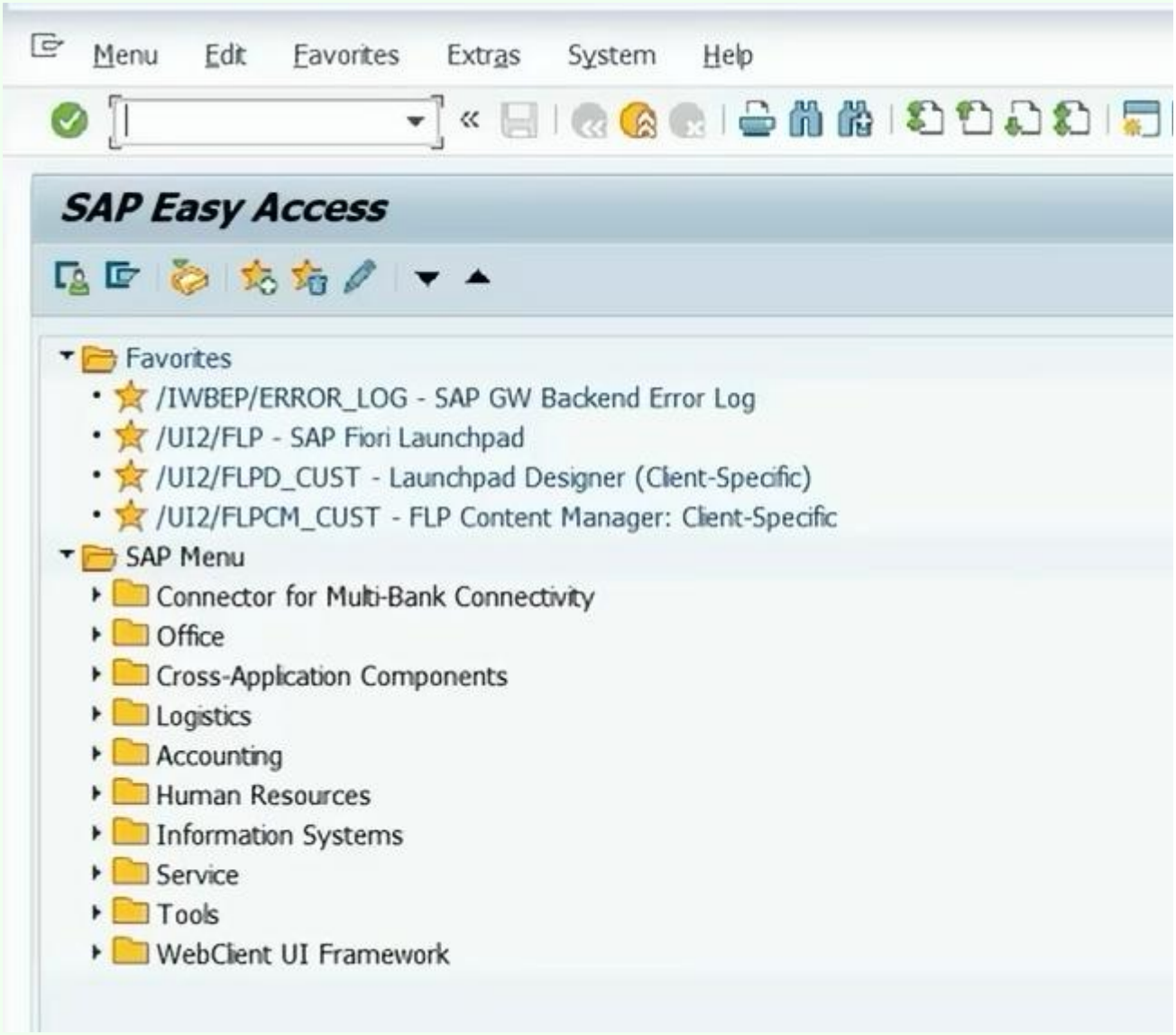
2.3 SAP Organizational Hierarchy — Overview

The SAP enterprise structure for ZenithPharma is a multi-level hierarchy that maps every real-world organizational unit to a corresponding SAP object. The hierarchy flows as follows: Client → Company Code → Business Area → Plants → Storage Locations (MM side) and Client → Company Code → Sales Organization → Distribution Channel →

Division → Sales Area (SD side). All financial postings from MM and SD roll up automatically into the FI general ledger.

[SCREENSHOT — SAP Logon Screen / SAP Easy Access Menu]

T-Code / Path: SAP GUI → Open SAP Logon → Log in to Client 200



3. Financial Accounting (FI) Module — Configuration

3.1 Module Overview

The Financial Accounting (FI) module is the financial backbone of ZenithPharma's SAP system. It records every business transaction in real time and produces statutory financial statements — Balance Sheet, Profit & Loss Account, and Cash Flow Statement — in compliance with the Companies Act, 2013, and Indian Accounting Standards (Ind AS). The FI module for ZenithPharma is configured with a focus on pharma-specific requirements: batch-level cost tracking, excise and GST compliance, and strict inter-company reconciliation between the two manufacturing plants.

3.2 Client Configuration

In SAP, the Client is the top-most organizational unit. All configuration, master data, and transactional data exist within a client. ZenithPharma operates on a dedicated production client.

SAP Parameter	Configured Value	Description
Client ID	200	ZenithPharma production client
Client Name	ZPIL SAP Live	Live production environment
Default Currency	INR	Indian Rupee
Default Language	EN	English

3.3 Company Code

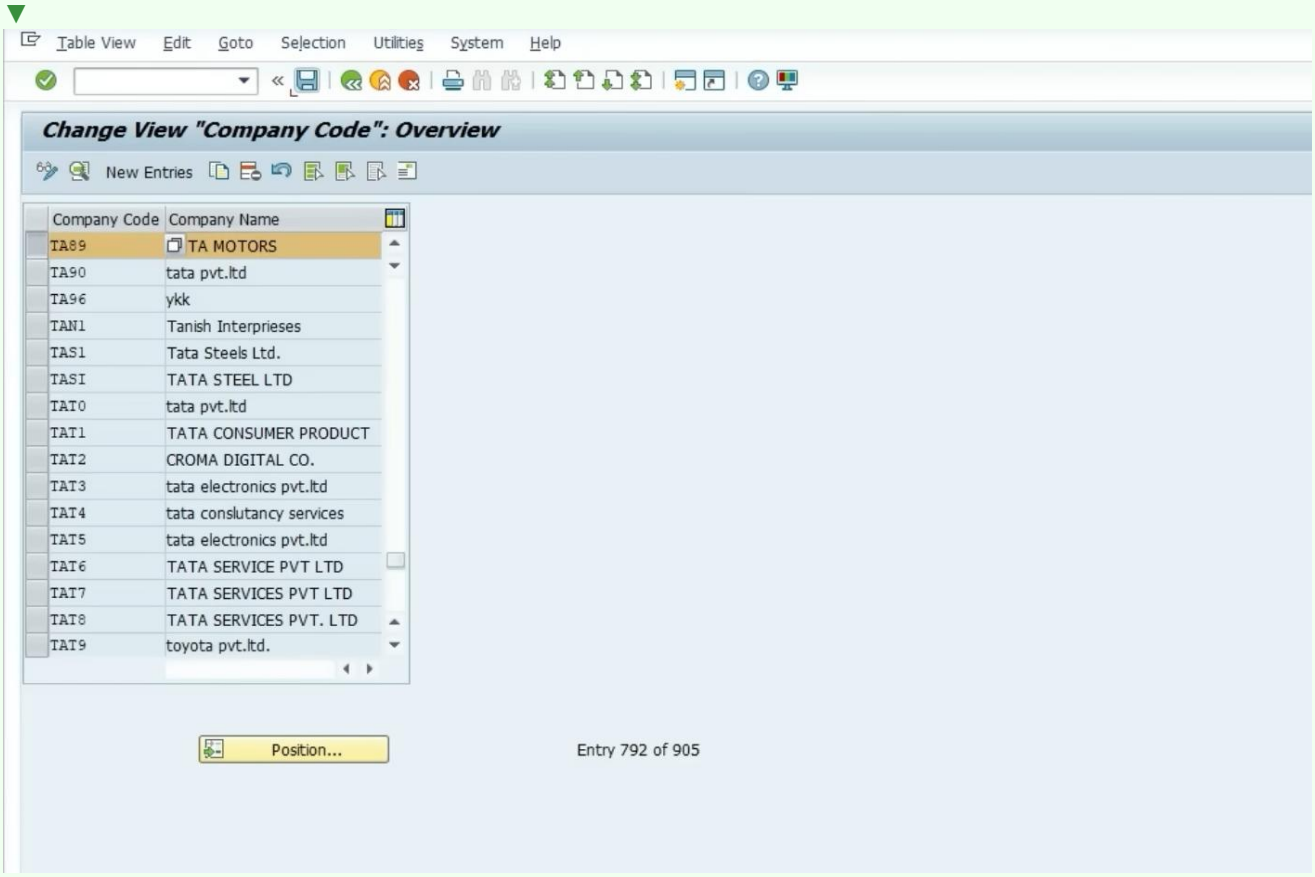
The Company Code is the smallest SAP organizational unit for which a complete, self-contained set of accounts can be maintained. It corresponds to a legal entity for which statutory financial statements are prepared. ZenithPharma has one Company Code representing its registered Indian entity.

Parameter	Value
Company Code	ZPIL
Company Name	ZenithPharma India Pvt. Ltd.
Street / City	Plot No. 47, Genome Valley, Hyderabad
State	Telangana
Country	India (IN)
Currency	INR

Parameter	Value
Language	EN (English)
Chart of Accounts	ZPCA
Fiscal Year Variant	ZV (April–March, 12+4 periods)
Posting Period Variant	ZP01
Credit Control Area	CC01

[SCREENSHOT — Company Code Configuration Screen]

T-Code / Path: SPRO → Enterprise Structure → Definition → Financial Accounting → Edit Company Code (OX02)



3.4 Chart of Accounts (COA)

ZenithPharma uses a custom Chart of Accounts named ZPCA, structured in alignment with Schedule III of the Companies Act, 2013 and pharma industry accounting norms. The COA defines all G/L account numbers used across both plants.

Account Range	Category	Account Type	Description
100000–149999	Fixed Assets	Balance Sheet	Plant, Machinery, Lab Equipment, Vehicles
150000–199999	Current Assets	Balance Sheet	Inventory, Cash, Bank, Receivables
200000–249999	Liabilities	Balance Sheet	Vendor Payables, Loans, Provisions
250000–299999	Equity	Balance Sheet	Share Capital, Reserves & Surplus
300000–349999	Revenue	P&L	Product Sales, Export Revenue
350000–399999	COGS	P&L	Raw Material Consumption, Batch Costs
400000–449999	Operating Exp	P&L	R&D, Regulatory, QA/QC Expenses
450000–499999	Tax Accounts	P&L	GST, TDS, Deferred Tax

3.5 Fiscal Year & Posting Period Configuration

ZenithPharma follows the Indian financial year — April 1 to March 31. The fiscal year variant ZV is configured with 12 regular posting periods (April to March) and 4 special periods reserved for year-end audit adjustments and closing entries.

Period No.	Month	Opening Date	Closing Date	Status
1	April	01.04.2025	30.04.2025	Closed
2	May	01.05.2025	31.05.2025	Closed
3	June	01.06.2025	30.06.2025	Closed
4	July	01.07.2025	31.07.2025	Closed
5	August	01.08.2025	31.08.2025	Closed
6	September	01.09.2025	30.09.2025	Closed
7	October	01.10.2025	31.10.2025	Closed
8	November	01.11.2025	30.11.2025	Closed
9	December	01.12.2025	31.12.2025	Closed
10	January	01.01.2026	31.01.2026	Closed
11	February	01.02.2026	28.02.2026	Open
12	March	01.03.2026	31.03.2026	Open —

Period No.	Month	Opening Date	Closing Date	Status
				Current
13–16	Special Periods	Year-End Only	Year-End Only	Open during closing

3.6 Document Types & Number Ranges

SAP uses Document Types to classify financial transactions. Each document type is assigned a number range, which controls how document numbers are generated. ZenithPharma's document types are configured as follows:

Doc Type	Code	Description	Number Range	Account Types Allowed
Vendor Invoice	KR	Pharma supplier invoices	5100000000–5199999999	Vendor + G/L
Customer Invoice	DR	Sales invoices to distributors	5200000000–5299999999	Customer + G/L
GL Journal Entry	SA	Internal adjustment entries	5000000000–5099999999	G/L only
Outgoing Payment	ZP	Payment to vendors	5300000000–5399999999	Vendor + Bank
Incoming Payment	ZR	Collections from distributors	5400000000–5499999999	Customer + Bank
Credit Memo	KG	Vendor credit / returns	5500000000–5599999999	Vendor + G/L
Asset Acquisition	AA	Lab/plant equipment purchases	5600000000–5699999999	Asset + G/L

3.7 Key FI Customization Steps

Step 1 — Define Company

Navigation Path: SPRO → IMG → Enterprise Structure → Definition → Financial Accounting → Define Company

- Company ID: ZPIL | Name: ZenithPharma India Pvt. Ltd.
- Country: IN | Currency: INR | Language: EN
- T-Code reference: OX15

Step 2 — Create and Configure Company Code

Navigation Path: SPRO → Enterprise Structure → Definition → Financial Accounting → Edit, Copy, Delete, Check Company Code

- Company Code ZPIL created with all parameters as per Section 3.3
- T-Code: OX02

Step 3 — Assign Company Code to Chart of Accounts

Navigation Path: SPRO → Financial Accounting → General Ledger → G/L Accounts → Master Data → Assign Company Code to Chart of Accounts

- ZPIL → ZPCA | T-Code: OB62

Step 4 — Define and Assign Fiscal Year Variant

Navigation Path: SPRO → Financial Accounting → Financial Accounting Global Settings → Fiscal Year → Maintain Fiscal Year Variant

- Variant ZV: April (Period 1) to March (Period 12) + 4 Special Periods
- Assignment: T-Code OB37 — Assign ZV to Company Code ZPIL

Step 5 — Configure Field Status Variants

Field Status Variants control which fields are mandatory, optional, or suppressed during document entry. ZenithPharma requires batch fields and profit center fields to be mandatory for all inventory-related G/L accounts.

- Field Status Variant: ZFS1 | Assigned to Company Code ZPIL via T-Code OBC4

Step 6 — Define Tax Procedure (TAXINN)

India uses the TAXINN tax procedure for GST calculations. ZenithPharma has the following GST rates configured for pharma products:

Product Category	HSN Code Range	GST Rate	Condition Type
Essential Medicines (DPCO)	3004	Nil / 5%	JIVP
OTC Formulations	3004	12%	JIVC
API / Bulk Drugs	2941	12%	JIVP
Medical Devices / Equipment	9018	12%	JIVC
Nutraceuticals / Vitamins	2106	18%	JIVC

4. Materials Management (MM) Module — Configuration

4.1 Module Overview

The Materials Management module handles all procurement and inventory management activities for ZenithPharma. Given the pharmaceutical industry's strict regulatory requirements, the MM module is configured with batch management, shelf-life tracking, quality inspection at goods receipt, and GMP-compliant storage location management. The module covers the complete Procure-to-Pay cycle from purchase requisition through goods receipt, invoice verification, and vendor payment.

4.2 Plant Configuration

A Plant in SAP represents a manufacturing facility, warehouse, or distribution center. ZenithPharma operates two GMP-certified plants:

Plant Code	Plant Name	Location	Type	Capacity
ZP01	ZenithPharma Hyderabad Plant	Hyderabad, TG	Formulation Manufacturing	500M tablets/year
ZP02	ZenithPharma Ahmedabad Plant	Ahmedabad, GJ	API & Injectable Mfg.	200M units/year

4.3 Storage Locations

Storage Locations represent physically separate storage areas within a plant. For a pharmaceutical company, these must align with GMP requirements — separate areas for quarantine, approved, rejected, and controlled substances.

Plant	Stor. Loc Code	Storage Location Name	GMP Zone
ZP01	RM01	Raw Material Store — General	Zone A
ZP01	RM02	Raw Material Store — Controlled Substances	Zone B (Restricted)
ZP01	QA01	Quarantine Store (Pre-QC approval)	Zone C
ZP01	FG01	Finished Goods Store — Approved	Zone A
ZP01	RJ01	Rejected Material Store	Zone D
ZP01	WIP1	Work-In-Progress / Dispensing Area	Zone A
ZP02	RM03	Raw Material Store — API Plant	Zone A

Plant	Stor. Loc Code	Storage Location Name	GMP Zone
ZP02	FG02	Finished Goods — API / Injectable	Zone A
ZP02	QA02	Quarantine Store — Ahmedabad	Zone C

4.4 Purchase Organization

The Purchase Organization is responsible for all procurement activities and legal purchase agreements. ZenithPharma uses a centralized procurement model where a single Purchase Organization handles all purchasing for both plants. This enables consolidated vendor negotiations and volume-based contracts.

Parameter	Value
Purchase Org Code	ZP_PO
Purchase Org Name	ZenithPharma Central Procurement
Scope	Cross-Plant — Covers ZP01 and ZP02
Assigned Company Code	ZPIL
Currency	INR
Procurement Type	Domestic + Import (API, excipients)

4.5 Purchase Groups

Purchase Group	Code	Responsible For	Approx. Annual Spend
Active Pharma Ingredients	PG01	APIs, bulk drugs, intermediates	INR 180 Cr
Excipients & Packaging	PG02	Excipients, blister packs, labels	INR 95 Cr
Lab & QC Materials	PG03	QC reagents, testing kits, media	INR 22 Cr
Engineering & Spares	PG04	Plant equipment, spare parts	INR 18 Cr
IT & Services	PG05	IT, AMC, contract services	INR 12 Cr

4.6 Material Types

Material Type	SAP Code	Description	Val. Method

Material Type	SAP Code	Description	Val. Method
Active Ingredients (API)	ROH	Raw APIs — core drug substances	Moving Avg.
Excipients	ROH	Binders, fillers, coatings	Moving Avg.
Packaging Materials	VERP	Primary & secondary packaging	Moving Avg.
Semi-Finished Goods	HALB	Granules, intermediate batches	Standard Price
Finished Formulations	FERT	Tablets, capsules, syrups, injectables	Standard Price
Lab Consumables	NLAG	Non-stock, expensed on consumption	N/A
Spare Parts	ERSA	Plant maintenance spares	Moving Avg.

4.7 Batch Management Configuration

Batch management is mandatory in pharmaceutical manufacturing for regulatory traceability. Every raw material and finished product must be tracked at batch level — from vendor's batch number through production batch to customer delivery. ZenithPharma configures batch management at plant level.

- Batch Level: Plant Level (each plant maintains independent batch records)
- Batch Number Format: ZP-YYYYMM-XXXXX (e.g., ZP-202604-00123)
- Shelf-Life Management: Minimum remaining shelf life enforced at GR — T-Code MIGO
- Batch Classification: Batch characteristics configured for potency, expiry date, vendor batch, COA reference
- Batch Status Management: Restricted batches blocked automatically until QA releases — T-Code MSC2N

Note: Batch management is a pharma-specific MM configuration not typically required in other industries. This is one of the most critical compliance elements of this blueprint.

4.8 Vendor Account Groups

Account Group	Code	Description	Partner Function
Domestic API Vendors	DAPI	Indian API manufacturers / suppliers	LF (Vendor)
Import Vendors	IMPV	International raw material suppliers	LF (Vendor)
Packaging Vendors	PKGV	Packaging material suppliers	LF (Vendor)
Service Providers	SRVV	AMC, calibration, contract	LF (Vendor)

Account Group	Code	Description	Partner Function
		services	
One-Time Vendors	ONEV	Petty purchase / emergency vendors	LF (Vendor)

4.9 Key MM Customization Steps

Step 1 — Define Plants and Assign to Company Code

Path: SPRO → Enterprise Structure → Definition → Logistics General → Define Plant

- Create ZP01 (Hyderabad) and ZP02 (Ahmedabad) | T-Code: OX10
- Assign both plants to Company Code ZPIL | T-Code: OX18

Step 2 — Maintain Storage Locations

Path: SPRO → Enterprise Structure → Definition → Materials Management → Maintain Storage Location

- All 9 storage locations created across ZP01 and ZP02 | T-Code: OX09

Step 3 — Define and Assign Purchase Organization

- Create ZP_PO | T-Code: OX08
- Assign ZP_PO to Company Code ZPIL | T-Code: OX01
- Assign ZP_PO to Plants ZP01 and ZP02 | T-Code: OX17

Step 4 — Configure Material Master Views

Material Master records in SAP contain multiple views (organizational levels). For pharma, the following views are mandatory:

View	Tab in MM01	Key Fields for Pharma
Basic Data 1	Basic Data	Material desc., base UoM, HSN code
Purchasing	Purchasing	Purchase group, order unit, GR processing time
MRP 1/2	MRP	MRP type, lot size, reorder point
Quality Management	QM	QM procurement active, inspection type 01
Warehouse Mgmt.	Warehouse	Storage type, picking strategy
Batch Classification	Classification	Batch characteristics: potency, expiry, COA
Accounting 1	Accounting	Valuation class, price control (S/V), std. price

Step 5 — Configure Valuation and Account Determination

ZenithPharma uses the following valuation classes to map materials to G/L accounts automatically:

Valuation Class	Material Category	G/L Account (Stock)	G/L Account (Consumption)
3000	APIs and Bulk Drugs	150100	350100
3001	Excipients	150200	350200
3002	Packaging Materials	150300	350300
7900	Finished Formulations	155000	355000
7901	Semi-Finished / WIP	156000	356000

5. Sales & Distribution (SD) Module — Configuration

5.1 Module Overview

The Sales & Distribution module manages the complete Order-to-Cash cycle for ZenithPharma — from customer inquiry and drug price negotiations through sales order processing, outbound delivery, batch determination, billing, and accounts receivable collection. The SD module is configured with pharma-specific requirements: DPCO price controls, batch-level delivery tracking, export documentation support, and channel-wise pricing strategies for different customer segments.

5.2 Sales Organization

Parameter	Value
Sales Org Code	ZP_SO
Sales Org Name	ZenithPharma National Sales
Company Code	ZPIL
Currency	INR
Scope	PAN India + Export Markets
Sales Calendar	Indian business calendar (excl. national holidays)

5.3 Distribution Channels

Distribution Channels define the pathway through which ZenithPharma's products reach the end customer. Different channels have different pricing strategies, credit terms, and discount structures.

Channel Code	Channel Name	Customer Type	Pricing Strategy
WS	Wholesale Distributors	C&F agents, Super stockists	MRP minus scheme discount
HP	Hospital & Institution	Government hospitals, nursing homes	Tender pricing / DPCO rate
RT	Retail Pharmacy	Individual chemist outlets	Distributor landed price
EX	Export / International	Foreign buyers, NGOs	FOB/CIF USD pricing

5.4 Divisions

Division Code	Division Name	Product Lines
FO	Formulations Division	Tablets, Capsules, Syrups, Ointments
IN	Injectables Division	IV Fluids, Ampoules, Vials
AP	API / Bulk Drugs Division	Active Pharmaceutical Ingredients
NT	Nutraceuticals Division	Vitamins, supplements, health foods

5.5 Sales Areas

The Sales Area is the unique combination of Sales Organization + Distribution Channel + Division. ZenithPharma configures 8 Sales Areas to cover all meaningful business scenarios:

Sales Org	Channel	Division	Sales Area Description
ZP_SO	WS	FO	Wholesale — Formulations (Domestic)
ZP_SO	WS	IN	Wholesale — Injectables (Domestic)
ZP_SO	HP	FO	Hospital Segment — Formulations
ZP_SO	HP	IN	Hospital Segment — Injectables
ZP_SO	RT	FO	Retail Pharmacy — Formulations
ZP_SO	EX	FO	Export — Formulations
ZP_SO	EX	IN	Export — Injectables
ZP_SO	WS	AP	Wholesale — API / Bulk Drugs

5.6 Shipping Points & Routes

Shipping Point	Code	Plant	Dispatch Mode
Hyderabad Dispatch Hub	SPHYD	ZP01	Road (National courier + own fleet)
Ahmedabad Dispatch Hub	SPAHM	ZP02	Road + Rail for bulk API
Export Hub — Hyderabad	SPEXP	ZP01	Air Cargo (IATA) + Sea Freight

5.7 Customer Account Groups

Account Group	Code	Description	Credit Mgmt.
Wholesale Distributor	WSCU	C&F Agents, Super Stockists	Credit limit enforced
Hospital / Institution	HOSP	Govt. & Private Hospitals	Government — extended 90 days
Retail Pharmacy	RTPH	Individual chemist shops	Cash / COD default
Export Customer	EXCU	International buyers	LC / advance payment
One-Time Customer	OTCU	Spot / emergency sales	No credit — advance only

5.8 Pricing Procedure — ZPRIC

ZenithPharma uses a custom pricing procedure named ZPRIC. This procedure calculates the final price for each sales document by applying a sequence of condition types in a specific order. The pharma industry has unique pricing requirements — MRP is a legally mandated maximum price (under DPCO), and all discounts are calculated from MRP downward.

Step	Cond. Type	Description	From	Calculation
10	ZMRP	Maximum Retail Price (MRP — DPCO)	—	Manual entry — legally mandated
20	ZSD1	Trade Scheme / Primary Discount	10	% on MRP
30	ZSD2	Secondary Distributor Discount	10	% on MRP
40	ZCD1	Cash Discount (if applicable)	10	% on MRP

Step	Cond. Type	Description	From	Calculation
50	ZNPV	Net Price to Distributor (NPD)	—	= Step 10 − 20 − 30 − 40
60	JIVP	GST — CGST (2.5% / 6%)	50	% on Net Price
70	JIVC	GST — SGST/IGST (2.5%/6%/12%/18%)	50	% on Net Price
80	ZFRT	Freight Charges	—	Manual / condition record
90	NETWR	Final Billing Value	—	= 50 + 60 + 70 + 80

Note: DPCO (Drug Price Control Order) caps the MRP of essential medicines listed in the National List of Essential Medicines (NLEM). ZenithPharma's pricing procedure is configured to enforce DPCO compliance — the system will not allow billing above the DPCO-notified price for scheduled formulations.

5.9 Key SD Customization Steps

Step 1 — Define Sales Organization and Assign to Company Code

- Create ZP_SO | T-Code: OVX5
- Assign ZP_SO to ZPIL | T-Code: OVX3

Step 2 — Define Distribution Channels and Divisions

- Create WS, HP, RT, EX channels | T-Code: OVX1
- Create FO, IN, AP, NT divisions | T-Code: OVX2
- Assign channels to Sales Org | T-Code: OVX4
- Assign divisions to Sales Org | T-Code: OVX5

Step 3 — Set Up Sales Areas

- Define all 8 Sales Area combinations in Customizing
- Path: SPRO → SD → Sales → Sales Documents → Define Sales Area

Step 4 — Configure Sales Document Types

Document Type	Code	Usage
Standard Order	ZOR	Regular sales orders to distributors
Rush Order	ZRO	Emergency/urgent orders
Return Order	ZRE	Customer returns / expired stock recall
Export Order	ZEX	Orders for international customers
Consignment Fill	ZKB	Stock placed at distributor on consignment

Step 5 — Configure Output Types for Billing

Output types define what documents are automatically generated after billing. ZenithPharma configures:

- RD00 — Invoice printout (standard SAP)
- ZGS1 — GST-compliant e-Invoice output (mandatory for turnover > INR 5 Cr)
- ZEX1 — Export invoice with shipping bill reference

6. Cross-Module Integration — FI, MM & SD

6.1 Overview

One of SAP's most powerful capabilities is real-time, automatic integration between all modules. Every procurement and sales transaction in ZenithPharma automatically generates corresponding financial postings in the General Ledger — without any manual journal entries. This section documents the exact accounting entries triggered by each key business event.

6.2 MM–FI Integration: Procure-to-Pay Accounting Flow

Business Event	T-Code	Debit Account	Credit Account	Amount Basis
Purchase Order Created	ME21N	No FI posting (commitment only)	—	PO Value
Goods Receipt (GR)	MIGO	Stock Account (150100)	GR/IR Clearing (200500)	PO Price × Qty
Quality Release (QM)	QA32	No FI posting	—	—
Invoice Verification	MIRO	GR/IR Clearing (200500)	Vendor Payable (200100)	Invoice Value
GST Input Tax Credit	MIRO	GST ITC Account (450100)	Vendor Payable (200100)	GST Amount
Vendor Payment	F110	Vendor Payable (200100)	Bank Account (150900)	Payment Amount
TDS Deduction	F110	Vendor Payable (200100)	TDS Payable (450200)	TDS Amount

6.3 SD–FI Integration: Order-to-Cash Accounting Flow

Business Event	T-Code	Debit Account	Credit Account	Amount
Sales Order Created	VA01	No FI posting	—	—
Outbound Delivery	VL01N	No FI posting	—	—
Goods Issue (Delivery)	VL02N	COGS — Finished Goods(355000)	FG Stock Account (155000)	Std. Cost
Customer Billing	VF01	Customer Receivable (150700)	Revenue Account (300100)	Billing Value
GST Output Tax	VF01	Customer Receivable (150700)	GST Output Payable(450300)	GST Amount

Business Event	T-Code	Debit Account	Credit Account	Amount
Customer Payment	F-28	Bank Account (150900)	Customer Receivable(150700)	Amt. Received
Credit Memo / Return	VF01	Revenue Account (300100)	Customer Receivable(150700)	Credit Amt.

6.4 MM–SD Integration: Batch Traceability

In pharmaceuticals, end-to-end batch traceability is a regulatory requirement. SAP provides this through the Batch Where-Used List (MB56). When a finished drug batch is delivered to a customer, the system can trace back to the exact raw material batches used in its manufacture. ZenithPharma's batch traceability chain is:

- Vendor Batch (GR via MIGO) → ZenithPharma internal batch number assigned
- Dispensing (to production) → WIP batch created with linkage to RM batches
- Manufacturing Order completion → Finished goods batch created, linked to WIP
- Sales Delivery (VL01N) → Finished goods batch assigned to delivery line
- Full trace report available: T-Code MB56 — Batch Where-Used List

[SCREENSHOT — SAP SPRO — IMG Configuration Tree]

T-Code / Path: SAP GUI → T-Code: SPRO → 'SAP Reference IMG'

Implementation Activities
Edit
Goto
Additional Information
Utilities
System
Display IMG
Existing BC Sets
BC Sets for Activity
Activated BC Sets for Activity
Structure
SAP Customizing Implementation Guide
Commercial Project Management
Activate Business Functions
Conversion of Accounting to SAP S/4HANA
ABAP Platform
Enterprise Structure
Cross-Application Components
Mobile Application Integration Framework Configuration
SAP Portfolio and Project Management
Financial Accounting
Financial Supply Chain Management
Multi-Bank Connectivity Connector
Strategic Enterprise Management/Business Analytics
Controlling
Planning and Consolidation
SAP S/4HANA for Group Reporting
Investment Management
SAP Global Trade Services, edition for SAP HANA
Enterprise Controlling
Flexible Real Estate Management (RE-FX)
Logistics - General
Environment, Health and Safety
Product Safety and Stewardship
Product Compliance
Sales and Distribution
Materials Management
Governance, Risk and Compliance
Logistics Execution
SCM Extended Warehouse Management
Transportation Management
Quality Management
Plant Maintenance and Customer Service
Customer Service
Production

7. Tech Stack & Complete T-Code Reference

7.1 SAP Components Utilized

Component	Version / Type	Purpose in ZenithPharma
SAP ECC 6.0	EhP 8	Core ERP — all modules
SAP FI	Financial Accounting	GL, AP, AR, Asset Accounting
SAP CO	Controlling	Cost Center, Profit Center, Product Costing
SAP MM	Materials Management	Procurement, Inventory, Batch Mgmt.
SAP QM	Quality Management	Inspection lots, usage decisions, COA
SAP SD	Sales & Distribution	Order-to-Cash, pricing, billing
SAP WM	Warehouse Management	GMP-compliant storage type management
SAP SPRO / IMG	Configuration Tool	All customization activities
SAP GUI 7.60	Front-End Client	User interface for all transactions
SAP Solution Manager	7.2	Project management, test cases, monitoring

7.2 Complete Transaction Code (T-Code) Reference

T-Code	Module	Transaction Name	Used For
OX15	FI	Define Company	Create company entity
OX02	FI	Edit Company Code	Configure Company Code ZPIL
OB62	FI	Assign COA to Company Code	Link ZPCA to ZPIL
OB29	FI	Maintain Fiscal Year Variant	Define ZV (Apr–Mar)
OB37	FI	Assign FY Variant to Co. Code	Assign ZV to ZPIL
OBC4	FI	Field Status Variants	Define mandatory fields
FB50	FI	Enter GL Account Document	Manual journal entries
F110	FI	Automatic Payment Run	Vendor payment processing
F-28	FI	Post Incoming Payments	Customer payment posting

T-Code	Module	Transaction Name	Used For
FS10N	FI	G/L Account Balance	View GL balances
OX10	MM	Define Plant	Create ZP01, ZP02
OX18	MM	Assign Plant to Company Code	ZP01/02 → ZPIL
OX09	MM	Maintain Storage Locations	Create all storage locations
OX08	MM	Define Purchase Organization	Create ZP_PO
OX17	MM	Assign PurchOrg to Plant	ZP_PO → ZP01, ZP02
MM01	MM	Create Material Master	Create new material records
MM03	MM	Display Material	View material master data
ME21N	MM	Create Purchase Order	Raise PO to vendor
ME23N	MM	Display Purchase Order	View PO details
MIGO	MM	Goods Movement	GR, GI, transfer postings
MIRO	MM	Invoice Verification (LIV)	Post vendor invoices
MB52	MM	Warehouse Stocks	View stock by storage location
MB56	MM	Batch Where-Used List	Pharma batch traceability
MSC2N	MM	Change Batch Master	Change batch status (QA release)
OVX5	SD	Define Sales Organization	Create ZP_SO
OVX3	SD	Assign Sales Org to Co. Code	ZP_SO → ZPIL
OVX1	SD	Define Distribution Channel	Create WS, HP, RT, EX
OVX2	SD	Define Division	Create FO, IN, AP, NT
OVX4	SD	Assign Channel to Sales Org	All channels → ZP_SO
VA01	SD	Create Sales Order	Enter new customer order
VA03	SD	Display Sales Order	View order details
VL01N	SD	Create Outbound Delivery	Pick and pack for shipment
VF01	SD	Create Billing Document	Generate invoice / credit memo
VK11	SD	Create Condition Records	Maintain pricing/discount records

8. Unique Features & Highlights of This Blueprint

8.1 What Distinguishes This Configuration

- Pharmaceutical-Specific Batch Management: Unlike generic SAP implementations, this blueprint includes full batch management configuration with shelf-life tracking, GMP zone classification, and QM inspection integration — a regulatory necessity under Schedule M of the Drugs & Cosmetics Act, 1940.
- DPCO-Compliant Pricing Procedure: The custom pricing procedure ZPRIC enforces Drug Price Control Order compliance — the system prevents billing above DPCO-notified MRP for scheduled essential medicines. This is unique to pharma implementations.
- GMP-Aligned Storage Location Design: Each storage location is mapped to a specific GMP zone (A/B/C/D) — quarantine zones prevent approved and unapproved material from mixing. This is a regulatory audit requirement under WHO-GMP guidelines.
- Multi-Channel Distribution with Channel-Specific Pricing: Four distribution channels (Wholesale, Hospital, Retail, Export) each carry distinct pricing strategies, credit terms, and discount structures — reflecting real-world pharma commercial operations.
- GST TAXINN Configuration with HSN-Level Tax Rates: Different pharma product categories attract different GST rates (Nil/5%/12%/18%). The blueprint maps each material type to its correct HSN code and GST condition type — essential for GST filing compliance.
- Centralized Cross-Plant Procurement: A single Purchase Organization (ZP_PO) serves both manufacturing plants, enabling consolidated vendor contracts, volume-based pricing, and unified vendor evaluation — a cost optimization strategy common in multi-plant pharma companies.
- Valuation Class — Material-to-Account Mapping: Five distinct valuation classes ensure that API costs, excipient costs, packaging costs, and finished goods costs flow into separate G/L accounts automatically — enabling product-line profitability analysis without manual allocation.

9. Future Improvements & Planned Enhancements

9.1 Short-Term Enhancements (0–6 Months)

- SAP QM (Quality Management) Full Configuration: Complete the Quality Management module integration — inspection plans, sampling procedures, usage decisions, and Certificate of Analysis (COA) generation for every finished goods batch. This is mandatory for export compliance.
- GST e-Invoicing Integration (IRP): Configure SAP SD to generate e-invoices via the Invoice Registration Portal (IRP) as mandated by GSTN for companies with turnover above INR 5 Crores. This includes IRN number generation and QR code printing on invoices.
- SAP MM Import Procurement Process: Configure complete import procurement cycle including Import Purchase Orders (in USD/EUR), customs duty condition types, Bill of Entry processing, and IGST on imports — required for API sourcing from China and Europe.

9.2 Medium-Term Enhancements (6–18 Months)

- SAP PP (Production Planning) Integration: Extend the blueprint into the Plan-to-Produce cycle — Bill of Materials (BOM) for drug formulations, Master Recipes (Process Orders for pharma), Work Centers (production lines), Batch Classification at production level, and integration with QM inspection lots.
- SAP S/4HANA Migration: Migrate the ECC 6.0 configuration to SAP S/4HANA. Key changes include: Universal Journal (ACDOCA merging FI + CO), Material Ledger activation for actual costing, SAP Fiori Launchpad replacing SAP GUI, and Business Partner (BP) replacing separate vendor/customer masters.
- SAP EWM (Extended Warehouse Management): Replace basic WM with Extended Warehouse Management for the Hyderabad plant — enabling temperature-zone controlled storage, automated pick-pack-ship workflows, and integration with RFID-based track-and-trace for cold chain products.

9.3 Long-Term Enhancements (18+ Months)

- SAP Track and Trace / Serialization: Implement serialization at saleable unit level to comply with DSCSA (US) and EU FMD regulations for export markets — assigning unique 2D barcodes to each pack for end-to-end supply chain visibility.
- SAP Analytics Cloud (SAC) Integration: Connect SAP S/4HANA data with SAP Analytics Cloud for real-time dashboards — plant-wise yield analysis, batch rejection trends, distributor performance scorecards, and DPCO compliance monitoring.
- Regulatory Document Management (SAP RIM): Implement SAP Regulatory Information Management for managing drug registration dossiers, license renewals, and country-wise regulatory submissions — critical for ZenithPharma's international expansion strategy.

10. Conclusion

This Capstone Project presents a detailed, industry-specific SAP configuration blueprint for ZenithPharma India Pvt. Ltd. — a fictitious pharmaceutical manufacturing and distribution company operating across Hyderabad and Ahmedabad. The blueprint covers all three mandated functional areas — Financial Accounting (FI), Materials Management (MM), and Sales & Distribution (SD) — with a level of depth and industry specificity that reflects real-world SAP implementation practice in the Indian pharmaceutical sector.

The FI configuration establishes a robust financial accounting foundation with Indian fiscal year compliance, GST-aligned tax procedures, pharma-specific document types, and a Chart of Accounts structured in accordance with Schedule III of the Companies Act, 2013. The MM configuration goes beyond standard procurement to address the unique demands of pharmaceutical manufacturing — batch management, shelf-life enforcement, GMP zone-based storage location design, QM-integrated goods receipt, and drug-specific material types with appropriate valuation classes. The SD configuration reflects the multi-channel commercial model of a pharma company — with DPCO-compliant pricing procedures, channel-wise discount structures, export documentation support, and GST e-invoicing readiness.

The cross-module integration documentation demonstrates a thorough understanding of how SAP's modules work together seamlessly — every procurement event triggers automatic FI postings, every billing event updates the General Ledger in real time, and every delivery creates a traceable audit trail from raw material batch to customer delivery. This level of integration eliminates manual reconciliation, reduces financial errors, and provides the real-time visibility that regulatory compliance in the pharma industry demands.

The future improvements roadmap reflects an awareness of the evolving SAP landscape — from GST e-invoicing compliance to S/4HANA migration, from QM integration to analytics — positioning ZenithPharma for sustainable growth and regulatory excellence.

— End of Report —

Submitted by: Tushar Raj Padhy | Roll No: 23052851
SAP Integration Developer | Batch: 2027_SAP(OE) | KIIT DU | April 2026